

Question 2 (20 Marks)

2.1 Consider the following Java program:

```
01 //Program 1
02 public enum Color {
03     RED("FF0000"), GREEN("00FF00"), BLUE("0000FF");
04
05     private String hexCode;
06
07     private Color(String hexCode) {
08         this.hexCode = hexCode;
09     }
10
11     public String getHexCode() {
12         return hexCode;
13     }
14     public static void main(String [] a) {
15         for (Color color : Color.values())
16             System.out.println((color.ordinal() + 1) + " " + color +
17                               " has code " + color.getHexCode());
18     }
19 }
20
```

Based on the program above that defines an enumerated type called **Color**, answer the following questions:

a) What is the significance of the values "FF0000", "00FF00", and "0000FF" that are passed as arguments to the constructor of each value in the **Color** enumerated type?

Initialize the Hexcode field of each enum constant

(1 mark)

b) What is the purpose of the **getHexCode()** method in the **Color** enumerated type?

act as an accessor to get the return value of the Hexcode.

(1 mark)

c) What will be the output of the program?

1 RED has code FF0000
2 GREEN has code 00FF00
3 BLUE has code 0000FF

(3 marks)

d) Write a Java statement in **main()** method that uses the **compareTo()** method to display the results of a comparison of the positions of **RED** and **BLUE** in the **Color** enumerated type. What will the output be?

System.out.println(Colour.RED.compareTo(Colour.BLUE));

(2 marks)

Output = -2

e) Consider adding the following statements in the **main()** method:

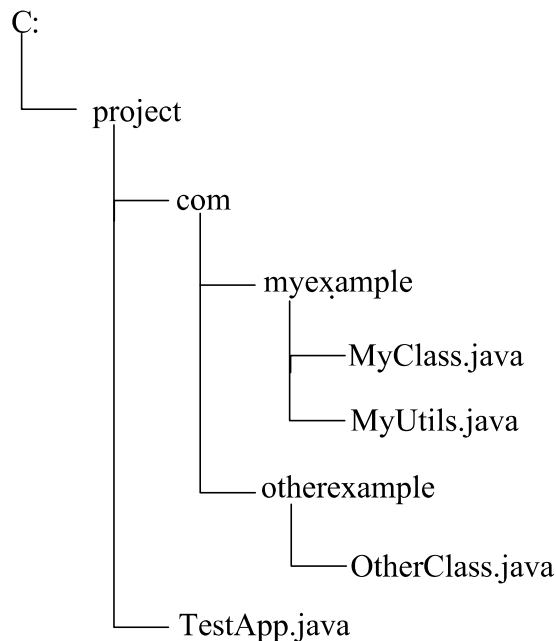
```
if (Color.GREEN.equals(Color.RED))
    System.out.println("GREEN and RED are two different colors.");
else
    System.out.println("GREEN and RED are the same color.");
```

The statements, however, include an error. What is the error, and how can it be fixed?

if (! Colour. GREEN. equals (Colour. RED))

(2 marks)

2.2 Given the directory structure as shown below:



a) What is the package name of **MyClass.java** and **MyUtils.java**?

com.myexample

(1 mark)

b) What is the package name of **OtherClass.java**?

com.otherexample

(1 mark)

c) In **TestApp.java**, how would you import all the classes in **com.myexample**?

import com.myexample.;*

(1 mark)

d) Suppose you want to move **MyUtils.java** to a new package called **com.myexample.util**. What changes would you need to make to the directory structure and the code?

package com.myexample.util

(3 marks)

2.3 Consider the Java program below and answer the following questions:

```
01 //Program 2
02 public class WrapperClass {
03     public static void main(String[] args) {
04         Integer num1 = 10;
05         Double num2 = 3.14;
06
07         //(a)Converting an object of a wrapper type to
08         //its corresponding primitive value
09         int value1 = num1.intValue();
10         double value2 = num2.doubleValue();
11
12         System.out.println("value1: " + value1);
13         System.out.println("value2: " + value2);
14
15         //(b)Adding value1 and value2 and storing the
16         //result in the result variable by converting
17         //it to the Double wrapper class
18         Double result = Double.valueOf(value1 + value2);
19
20         System.out.println("result: " + result);
21     }
22 }
```

- a) Fill in the blanks with an appropriate Java statement to convert an object of a wrapper type to its corresponding primitive value. **Note:** you must use a method provided in the wrapper class. What is the name of the conversion?

(3 marks)

- b) Fill in the blanks with an appropriate Java statement to add **value1** and **value2** and store the result in the **result** variable by converting it to the **Double** wrapper class. **Note:** you must use a method provided in the wrapper class.

(2 marks)